

The Process: A First-Class Analytical Unit for Orthogonal Information Extraction from Market Microstructure

Conditional-Information Signal Selection and the Effective
Degrees of Freedom of an Alpha Book

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Abstract

A high-frequency microstructure feed produces hundreds of engineered features, most of them mutually redundant. The operative research question is therefore not whether an individual feature predicts future price action, but how much *non-redundant* information the feature set carries in aggregate, and how many statistically *independent* bets that information supports. We formalize this question around a third first-class analytical unit—the *process*—placed beside the *feature* (what is computed) and the *algorithm* (how it trades). A process is a typed, registry-bound analytical contract that certifies whether a feature, or a derivative/combination of features, carries information about a precisely specified target. We then develop the mathematics the unit exists to serve. Adopting mutual information as the currency of prediction (justified by Fano’s inequality, which makes information a hard lower bound on achievable error), we define the effective degrees of freedom of a signal book through the fundamental law of active management, $IR \approx \bar{IC} \sqrt{N_{\text{eff}}}$ with $N_{\text{eff}} = N/[1 + (N - 1)\bar{\rho}]$, and show that the binding quantity is the *stress-regime* correlation, not the calm-regime correlation. We give five orthogonalization mechanisms, each as a process: conditional-mutual-information greedy selection, interaction-information synergy mining, training-prefix residualization, Marchenko–Pastur-denoised principal components, and a capstone decorrelated signal book that reports its own honest N_{eff} . We make “price action” explicit through an aspect×horizon target taxonomy, and we add a derivative-operator layer that mints new feature degrees of freedom (fractional differentiation, leaky integration, causal spectral features) under a program-level false-discovery ledger. The contribution is a unifying analytical contract together with the orthogonality-and-breadth mathematics for disciplined microstructure alpha discovery. This is a methods/framework paper: empirical estimates are specified as future work, and every unit ships behind a mandatory planted-signal conformance gate.

Keywords: market microstructure, mutual information, conditional mutual information, feature selection, effective breadth, fundamental law of active management, random matrix theory, false discovery rate, fractional differentiation.

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1 Introduction

1.1 Redundancy is the problem, not scarcity

A modern microstructure pipeline does not suffer from a shortage of candidate predictors. The platform underlying this work emits 236 engineered features every 100 milliseconds—order-book imbalance at several depths, trade-flow toxicity, realized-volatility estimators, entropy of book shape, funding pressure, liquidation geometry, and dozens of derived interactions. The difficulty is the opposite of scarcity: these features are heavily *co-determined*. Imbalance at level one and level two move together; a dozen volatility proxies share a common factor; flow and price are mechanically linked. Screening each feature in isolation and keeping “the ones with a high information coefficient” therefore answers the wrong question. It rewards a crowd of near-copies of a single underlying bet and quietly overstates how much independent evidence the research has actually accumulated.

The correct question has two parts, both of which this paper formalizes:

1. **Orthogonal extraction.** From a large, redundant feature set, isolate a basis of signals each of which carries information about future price action that the others do *not*. Redundancy must be measured in an information metric, not a linear one, because microstructure dependence is routinely nonlinear (a feature can be linearly uncorrelated with the target yet fully determine it—the canonical XOR obstruction).
2. **Effective degrees of freedom.** Quantify how many statistically independent bets the resulting book supports. This number—not the raw count of “significant” features—is what governs the achievable information ratio, and it is the quantity that collapses in exactly the market states where a portfolio most needs diversification.

1.2 Three first-class citizens

The platform answers these questions by promoting the analytical act itself to a typed, versioned, testable object. There are three first-class citizens, summarized in Table 1. The *feature* and the *algorithm* are familiar; the *process* is the unit this paper defines.

Citizen	Question it answers	Contract
feature	<i>what</i> is computed	ingestor columns / derived series
algorithm	<i>how</i> it trades	<code>MicrostructureAlgorithm</code>
process	<i>whether/where</i> information about price action exists	Process

Table 1: The three first-class citizens. A process formalizes what previously lived as scattered one-off analyses (a screener’s information coefficient, an information-theory engine’s mutual information, a spectral persistence study, a gradient-boosted importance test) behind one registry, one result schema, one command-line surface, and one persistence path.

1.3 Contributions

1. A formal, registry-bound *process contract* (Section 2): two kinds (evaluation and transform), a versioned result schema, and a typed-node refinement that treats the prediction target as its own first-class object.
2. The *information currency* (Section 3): why mutual information, not correlation, is the right dependence measure, anchored in Fano’s inequality and a cost-aware minimum-information gate.
3. The *effective degrees of freedom* of a signal book (Section 4): a derivation of $N_{\text{eff}} = N/[1 + (N - 1)\bar{\rho}]$ from the fundamental law of active management, and the calm-versus-stress decomposition that turns it into a risk disclosure.
4. Five *orthogonalization processes* (Section 5), each defined mathematically: conditional-MI greedy selection, interaction-information synergy mining, residualization, Marchenko–Pastur-denoised PCA, and the capstone signal book.
5. A *target taxonomy* (Section 6) making “price action” explicit along an aspect $\times$ horizon grid, and a *derivative-operator* layer (Section 7) that mints new feature degrees of freedom under a program-level false-discovery ledger.
6. An *estimation-hygiene* layer and the planted-signal verification discipline (Sections 8–9) that keep the degrees-of-freedom estimate honest.

This is deliberately a methods/framework paper. Where the contract is implemented we say so; where a mechanism is specified but not yet built we say so; and we report no fabricated empirical estimates. The one empirical fact we import as motivation—that a mid-price information coefficient of ≈ 0.45 collapses to ≈ 0.03 once conditioned on states where a passive fill would actually occur—is established in a companion study and cited, not re-derived here.

2 The Process as an Analytical Unit

2.1 Definition and the two kinds

Definition 2.1 (Process). A *process* is an analytical map—statistical, signal-processing, or machine-learning—that takes a panel of feature columns and a context, and certifies whether a feature, or a derivative/combination of features, carries information about a specified target of future price action. It is identified by a unique name, registered through a decorator, parameterized through a configuration section keyed by that name, and persisted through a single result schema.

Two kinds share the registry, command-line, and persistence surface.

Definition 2.2 (Evaluation and transform processes). An *evaluation process* is a map

$$\mathcal{E} : (\mathbf{D}, \mathcal{C}) \mapsto \mathcal{R},$$

where $\mathbf{D} \in \mathbb{R}^{T \times p}$ is a time-indexed feature panel, $\mathcal{C}$ is a *process context* (symbol, timeframe, price column, horizon set, cost parameters, target, auxiliary sources), and $\mathcal{R}$ is a *process result*: it scores existing columns and produces no new series. A *transform process* is a map

$$\mathcal{T} : (\mathbf{D}, \mathcal{C}) \mapsto (\mathbf{D}', \mathcal{R}), \quad \mathbf{D}' \in \mathbb{R}^{T \times m},$$

producing m new series $\mathbf{D}'$ on the input index (e.g. principal components, triple-barrier labels, residualized innovations). Derived series are themselves first-class evaluable inputs: the runner chains $\mathbf{D}'$ into any evaluation process, so $\mathcal{E} \circ \mathcal{T}$ is itself a process.

The closure of evaluation under transform is the engine of the whole framework: orthogonalization is a transform whose output is then scored by an evaluation process, so “extract the orthogonal part, then ask what it knows” is a single composed pipeline rather than two disconnected scripts.

2.2 The result contract

Definition 2.3 (Process result). A *process result* $\mathcal{R}$ (schema version 1) is the tuple

$$\mathcal{R} = (\text{id}, \text{kind}, \Pi, \mathcal{D}, \mathcal{P}, \mathcal{F}, \mathcal{S}, \mathcal{G}, \Sigma),$$

with run identifier id ; kind; parameters Π ; data fingerprint $\mathcal{D}$ (directory, dates, row/bar counts, content hash); provenance $\mathcal{P}$ (git SHA, dirty flag, timestamp); the set of tested features $\mathcal{F}$ and skipped features $\mathcal{S}$ with reasons; a list of findings $\mathcal{G}$; and a summary Σ . Each finding is

$$g = (\text{feature}, \text{horizon}, \text{metric}, \text{value}, \text{threshold}, p, p_{\text{adj}}, \text{informative}, \text{extras}),$$

where p_{adj} is the Benjamini–Hochberg-adjusted p -value within the run and *informative* is the boolean verdict $\text{value} \geq \text{threshold}$ (and p_{adj} below the false-discovery level where applicable).

The full result is written as an atomic JSON document; a queryable index row is mirrored to a local database on a best-effort basis. Two disciplines in this contract matter for the rest of the paper.

Remark 2.1 (The not-a-number partition, K2). Before scoring, every process partitions its columns through a guard that removes dead, constant, or too-thin columns—each skipped with a recorded reason (`all_nan`, `constant`, `n_valid<min`)—so a missing optional feature category never crashes a run and never silently contaminates an estimate. Information measures on a constant column are undefined, not zero; the partition makes that explicit rather than letting a degenerate estimator return a spurious value.

Remark 2.2 (False-discovery scope). Benjamini–Hochberg correction inside a result is *per run*. The index of all runs is a research log, *not* a meta-gate: aggregating “informative” flags across many runs inflates discoveries exactly as multiple testing predicts. The program-level ledger of Section 8 is the honest counterpart, and the deflated-Sharpe and reality-check gates remain authoritative for any promotion decision.

2.3 A typed-node view

The evaluation/transform dichotomy is a convenient first cut, but it already leaks: a portfolio-book transform performs construction rather than scoring, and a triple-barrier transform produces a *target* rather than a feature. The mature model is therefore a typed computation graph with four node kinds—*feature*, *transform*, *evaluation*, and *target*—rather than a two-element stack. We do not refactor preemptively; we record the target node as the first new kind, because making the prediction target explicit (Section 6) is the single highest-leverage change for honest information measurement.

3 Information as the Currency of Prediction

3.1 Why mutual information, not correlation

Let X be a feature and Y a target derived from future price action. The mutual information

$$I(X; Y) = \iint p(x, y) \log \frac{p(x, y)}{p(x)p(y)} dx dy = H(Y) - H(Y | X)$$

measures *any* statistical dependence, linear or not, and is invariant to monotone reparameterization of either variable. Correlation answers “how much, linearly”; mutual information answers “is there any exploitable dependence at all.” The reason this distinction is not cosmetic is a hard inequality.

Proposition 3.1 (Information lower-bounds achievable error). *Let $\hat{Y} = g(X)$ be any predictor of a discretized target Y taking values in a set of size K , and let $P_e = \Pr[\hat{Y} \neq Y]$. Then Fano’s inequality gives*

$$H(Y | X) \leq H_b(P_e) + P_e \log(K - 1),$$

hence

$$P_e \geq \frac{H(Y) - I(X; Y) - \log 2}{\log(K - 1)}.$$

The content of Proposition 3.1 is that no estimator, however clever, can drive the error below a floor set by $I(X; Y)$. Information is therefore not one diagnostic among several; it is the budget. A feature with negligible mutual information cannot be rescued by a better model, and a feature selection that maximizes information is maximizing the only quantity that bounds achievable prediction error (Cover & Thomas; the continuous analogue is the rate–distortion theorem).

3.2 The cost-aware information gate

Statistical information is necessary but not sufficient: a signal must carry enough information to pay the round-trip trading cost. The platform encodes this as a minimum-information threshold that every information-valued process compares against,

$$I_{\min} = I_{\min}(\text{fee}, \sigma_r, \text{kurt}(r)),$$

the least mutual information a feature must share with returns for a position to clear fees and slippage given the return scale and tail heaviness. A finding is *informative* only if its estimated information exceeds $I_{\min}$, not merely zero. This is the information-theoretic parent of the deflated, cost-adjusted gates used downstream, and it is why a process reports bits against a threshold rather than a bare p -value.

3.3 Estimating information without manufacturing it

Continuous mutual information is estimated with the Kraskov–Stögbauer–Grassberger k -nearest neighbour estimator. One practical caveat is load-bearing for everything that follows.

Remark 3.1 (Rank-transform before estimating). Raw-scale k -NN estimation manufactures a spurious information floor (empirically ≈ 0.07 bits on planted-null data) when feature and return scales differ, because the neighbour graph is dominated by the wider-scale coordinate. Rank/copula-transforming each input before estimation removes the artefact and makes the estimator scale-invariant. Every information-valued process in the framework rank-transforms its inputs; this single fix was found only because a planted-null test (Section 9) flagged a nonzero estimate where the true value was zero.

Directed dependence—which way information flows between two series—is measured by transfer entropy $\text{TE}(X \rightarrow Y) = \text{I}(Y_t; X_{t-1} | Y_{t-1})$ (Schreiber), with a linear (Gaussian) default where it is tolerable and the reverse direction $\text{TE}(Y \rightarrow X)$ retained as a control.

4 The Effective Degrees of Freedom of a Signal Book

The central quantity of the whole framework is not the information of any single feature but the *breadth* of the book of features that survive selection. We define it, derive it under the natural redundancy model, and then show why its calm-regime value is a dangerous overstatement.

4.1 The fundamental law and effective breadth

The fundamental law of active management (Grinold; Grinold & Kahn) relates the information ratio of a strategy to the skill per bet and the number of independent bets,

$$\text{IR} \approx \overline{\text{IC}} \sqrt{\text{BR}}, \tag{1}$$

where $\overline{\text{IC}}$ is the average information coefficient (skill) and BR is the breadth—the number of independent bets per period. The research programme is the pursuit of the two factors in (1): raise $\overline{\text{IC}}$ by finding real information, raise BR by combining *independent* pieces. The question is what BR equals when the pieces are correlated, which they always are.

Proposition 4.1 (Effective breadth under equicorrelation). *Let $s_1, \dots, s_N$ be standardized signals ($\mathbb{E}[s_i] = 0$, $\text{Var}(s_i) = 1$) with common pairwise correlation $\text{corr}(s_i, s_j) = \bar{\rho} \in [0, 1)$ for $i \neq j$. The equal-weight aggregate $S = \frac{1}{N} \sum_{i=1}^N s_i$ has variance*

$$\text{Var}(S) = \frac{1}{N^2} \left(N + N(N-1)\bar{\rho} \right) = \frac{1 + (N-1)\bar{\rho}}{N}.$$

Define the effective number of independent signals N_{eff} as the count of mutually independent unit-variance signals whose equal-weight aggregate has the same variance. Since n independent unit-variance signals give aggregate variance $1/n$, equating $1/N_{\text{eff}} = \text{Var}(S)$ yields

$$N_{\text{eff}} = \frac{N}{1 + (N-1)\bar{\rho}} \quad (2)$$

and, identifying breadth with effective independence ($\text{BR} = N_{\text{eff}}$) in (1),

$$\text{IR} \approx \overline{\text{IC}} \sqrt{\frac{N}{1 + (N-1)\bar{\rho}}}.$$

Proof. $\text{Var}(S) = N^{-2} [\sum_i \text{Var}(s_i) + \sum_{i \neq j} \text{cov}(s_i, s_j)] = N^{-2} [N + N(N-1)\bar{\rho}]$. Equate with the independent-case variance $1/N_{\text{eff}}$ and solve. $\square$

Corollary 4.1 (The two limits). *As $\bar{\rho} \rightarrow 0$, $N_{\text{eff}} \rightarrow N$: uncorrelated signals each count fully. As $\bar{\rho} \rightarrow 1$, $N_{\text{eff}} \rightarrow 1$: perfectly correlated signals count as one, regardless of how many were collected. Equation (2) is monotone decreasing in both N and $\bar{\rho}$ in the relevant regime, so adding a redundant signal raises N but cannot raise N_{eff} —the defining property a selection procedure must respect.*

Proposition 4.1 is the precise sense in which this paper “defines the degrees of freedom mathematically.” The raw count N of significant features is a vanity metric; N_{eff} is the quantity that enters the achievable information ratio, and the entire orthogonalization suite of Section 5 exists to make N_{eff} as close to N as the data permit.

4.2 The correlation that matters is the stress correlation

The correlation $\bar{\rho}$ in (2) is not a constant; it is regime-dependent, $\bar{\rho} = \bar{\rho}(\text{regime})$. Microstructure signals that are nearly uncorrelated in calm markets routinely co-move under stress, when a single liquidity factor drives every book at once. Hence two values of (2) must be reported:

$$N_{\text{eff}}^{\text{calm}} = \frac{N}{1 + (N-1)\bar{\rho}_{\text{calm}}}, \quad N_{\text{eff}}^{\text{stress}} = \frac{N}{1 + (N-1)\bar{\rho}_{\text{stress}}}, \quad \bar{\rho}_{\text{stress}} \geq \bar{\rho}_{\text{calm}}.$$

Remark 4.1 (Diversification evaporates when it is needed). A book with $\bar{\rho}_{\text{calm}} = 0.1$ but $\bar{\rho}_{\text{stress}} = 0.7$ has, for $N = 10$, $N_{\text{eff}}^{\text{calm}} \approx 5.3$ but $N_{\text{eff}}^{\text{stress}} \approx 1.4$: it offers five independent bets in quiet markets and barely more than one exactly when a drawdown is unfolding. The gap $N_{\text{eff}}^{\text{calm}} - N_{\text{eff}}^{\text{stress}}$ is therefore not a footnote—it *is* the risk disclosure of the book, and any honest report of breadth must state both numbers and the regime buckets they were computed in.

5 Extracting Orthogonal Information

We now give the five processes that build a high- N_{eff} book. Each is stated as a definition with its planted-signal contract; the layer they belong to is named in Table 2.

5.1 Conditional-MI greedy selection (the core extractor)

Definition 5.1 (Redundancy-aware information selection). Given a feature set $F = \{f_1, \dots, f_p\}$ and target r , build a selected set S_k greedily:

$$f_{(1)} = \arg \max_{f \in F} I(f; r), \quad f_{(k+1)} = \arg \max_{f \in F \setminus S_k} I(f; r | S_k),$$

terminating when the marginal conditional information gain falls below $\max(I_{\min}, \varepsilon)$. Report per selected feature the gain in bits and, per rejected runner-up, the redundancy $I(f; r) - I(f; r | S_k)$.

The conditional mutual information $I(f; r | S_k)$ is the information f shares with r that is *not already present* in the selected set. A feature that duplicates an already-selected one contributes ≈ 0 and is never chosen, so the selected set is orthogonal-by-construction in the information metric. This is strictly stronger than linear (principal-component) decorrelation, which a non-linear redundancy—two features that jointly but not marginally determine the target—slips past entirely.

Remark 5.1 (Relation to mRMR). The first-order expansion of Definition 5.1 is the minimum-redundancy maximum-relevance criterion, $I(f; r) - \frac{1}{|S_k|} \sum_{s \in S_k} I(f; s)$ (Peng, Long & Ding; unified by Brown et al.), and the lineage traces to Battiti’s mutual-information feature selection. The conditional form is the exact object the approximation targets.

Planted contract. On a frame containing a signal A , an exact noisy copy A' , and an independent signal B , the process must select $\{\text{one of } A, A'; B\}$ and never both copies; the cumulative-information curve must flatten after the second pick.

5.2 Interaction information: mining synergy

Definition 5.2 (Pairwise synergy). For features X, Y and target r , the interaction information is

$$\mathcal{S}(X, Y; r) = I(\{X, Y\}; r) - I(X; r) - I(Y; r).$$

$\mathcal{S} > 0$ signals *synergy* (the pair jointly carries more than the sum of its parts); $\mathcal{S} < 0$ signals *redundancy*.

Synergy is invisible to any single-feature screen and to Definition 5.1’s forward pass, because each marginal can be near zero while the joint is large. Run over the top- K features only ($K \approx 20 \Rightarrow 190$ pairs, the cost of k -NN estimation being the binding constraint), synergistic pairs become candidate engineered composites fed back as derived features (the partial information decomposition of Williams & Beer is the formal parent).

Planted contract. On an XOR construction $r = \text{sign}(x) \cdot \text{sign}(y)$, each marginal information is ≈ 0 while the joint exceeds the gate; the process must flag that pair and only that pair.

5.3 Residualization: pure-innovation extraction

Definition 5.3 (Residualization). For a target feature f and a conditioning set Z (a market mode, a cousin feature, or a principal component), the residual is

$$\text{res}_f(t) = f(t) - \beta^\top Z(t), \quad \beta = \arg \min_{\beta} \mathbb{E}_{\text{train}}[(f - \beta^\top Z)^2],$$

with β fit on the training prefix only. By the normal equations $\text{corr}(\text{res}_f, Z) = 0$ on the training sample, and approximately so out of sample; scoring res_f against r isolates what f knows that Z does not.

Residualization is the per-feature, tradeable analogue of the set-level selection of Definition 5.1: where conditional-MI selection *rank*s, residualization *emit*s a *series*. The training-prefix fit is the no-lookahead guarantee, verified by the same holdout perturbation test used for principal components.

5.4 Denoised principal components and the count of real factors

Linear orthogonalization by principal components is cheap and complementary, but on short samples most “components” are noise dressed as structure. Random matrix theory says exactly how much.

Proposition 5.1 (Marchenko–Pastur noise band). *Let $\mathbf{C}$ be the sample correlation matrix of a $T \times N$ panel of i.i.d. standardized noise with aspect ratio $q = N/T \in (0, 1]$. As $N, T \rightarrow \infty$ with q fixed, the eigenvalues of $\mathbf{C}$ fall in $[\lambda_-, \lambda_+]$ with*

$$\lambda_{\pm} = (1 \pm \sqrt{q})^2.$$

Eigenvalues exceeding λ_+ cannot be attributed to noise; the count

$$n_{\text{signal}} = \#\{i : \lambda_i > \lambda_+\}$$

estimates the number of genuine factors—the linear degrees of freedom—in the feature block.

Clipping or shrinking the eigenvalues inside $[\lambda_-, \lambda_+]$ denoises the correlation estimate (Laloux et al.; López de Prado), and n_{signal} is a direct, linear estimate of the degrees of freedom that Proposition 4.1 measures in the information metric. The principal-component transform reports n_{signal} and the maximum off-diagonal holdout correlation as its orthogonality witness.

Planted contract. A frame of three true factors plus forty noise features must return $n_{\text{signal}} \approx 3$, and the denoised holdout orthogonality must strictly improve on the raw.

5.5 The capstone: a decorrelated signal book with honest breadth

Definition 5.4 (Decorrelated signal book). Given features that pass selection (Definition 5.1), a stability gate, and a regime-robustness gate, the signal book is built by (i) sequentially residualizing in selection order so the surviving series are mutually orthogonal, (ii) walk-forward z -scoring them into `sig_*` columns, and (iii) reporting, as the book’s contract,

$$N_{\text{eff}} = \frac{N}{1 + (N - 1)\bar{\rho}}, \quad \text{IR}_{\text{expected}} = \overline{\text{IC}} \sqrt{N_{\text{eff}} \cdot \text{bars/year}},$$

computed at *both* calm and stress $\bar{\rho}$, together with the full provenance chain of every upstream run that admitted each entry.

The book is the object an algorithm consumes: an algorithm whose required inputs are the `sig_*` columns and whose step is risk-parity weighting. The division of labour is the thesis of the framework in one line: *processes certify what is combinable; algorithms combine.*

Process	Kind	Role in raising N_{eff}
<code>cmi_select</code>	evaluation	selects an information-orthogonal basis at the source
<code>interaction_synergy</code>	evaluation	recovers joint information no marginal screen sees
<code>lead_lag_te</code>	evaluation	cross-symbol directed edges—the cheapest decorrelation
<code>residualize</code>	transform	emits the pure innovation of a feature beyond a set
<code>pca_combo</code> + MP	transform	linear orthogonal composites; counts real factors
<code>signal_book</code>	transform	assembles the mutually orthogonal book; reports N_{eff}

Table 2: The orthogonalization suite. Each process either raises the effective breadth N_{eff} or certifies it. `pca_combo` is implemented; the others are specified (Section 9).

6 Defining the Target: What “Price Action” Means

Mutual information is always information *about something*. A pervasive but silent error is to let every process default to simple forward returns while claiming to measure “information about price action.” Price action is not one thing, and the literature has studied features against many distinct targets. We make the target a first-class node.

Definition 6.1 (Target node). A *target* is a map $\tau : (\text{prices, bars, horizon}) \mapsto \mathbb{R}^T$ with a declared kind (continuous, binary, categorical), resolved from a registry and carried in the process context, so that every process computes $I(f; \tau)$ against one explicit, named target rather than an implicit default. Closing this gap is the highest-leverage single change in the framework: it makes “mutual information between a feature and price action” honest about *which* price action.

Table 3 is the resulting design surface: a run is the quadruple (process $\times$ target $\times$ horizon $\times$ symbol).

Aspect	Definition	Literature / status
Direction	$\text{sign}(r_h)$	classification; partial support
Magnitude	$r_h, r_h $	rate-distortion; <i>shipped default</i>
Volatility	future realized variance over h	HAR (Corsi); a target, not a scaler
Path	triple-barrier label, time-to-touch	López de Prado; partial
Regime change	$\text{Pr}(\text{switch})$	Hamilton; unbuilt
Jump / tail	Lee–Mykland statistic	jump literature; algorithm only
Fill-conditional	move at the transacted price	Kyle/Stoikov; the missing bridge

Table 3: An aspect $\times$ horizon taxonomy of price-action targets (horizons: tick-second, minute-bar, multi-day). Two cells are both unbuilt and high value: *volatility-as-target* (a whole tradeable family the platform cannot currently screen for) and *fill-conditional move* —the target defined on the price one actually transacts at, where the mid-price edge of ≈ 0.45 is known to collapse to ≈ 0.03 . The latter is the execution-realism gap made measurable.

7 Minting Feature Degrees of Freedom

Orthogonalization raises N_{eff} toward the number of distinct bets present in the feature set; it cannot create a bet that no feature expresses. The complementary lever is a transform process, **feature_ops**, that applies operators to a chosen base-column set and emits new **op_*** columns—candidate new degrees of freedom—each chainable into the evaluation and selection processes above. Only the operators that are genuinely missing from the existing feature set are proposed; the high-value, not-yet-built ones are:

1. **Fractional differentiation** $(1 - B)^d$, $0 < d < 1$ (López de Prado, after the long-memory literature). Integer differencing destroys the memory that carries predictive content; fractional differencing makes a level series stationary while *preserving* memory, so downstream information and IC estimates are both honest (stationary) and not amputated. This is the single highest-value missing operator.

2. **Leaky / cumulative integration** $y_t = \varrho y_{t-1} + x_t$, accumulating a memoryless flow or imbalance into a persistent pressure state for longer-horizon targets.
3. **Causal spectral features**: rolling band-power, dominant frequency, and rolling spectral entropy, exposing oscillatory structure as a persisted feature rather than as a one-off analysis.
4. **Distribution-free normalization**: rolling rank/quantile and robust (median/MAD) z -scores, which both help the rank-based information estimator of Remark 3.1 and are fat-tail robust.
5. **Nonlinear bases**: polynomial/spline expansions and a sign $\times$ magnitude split, so that even linear processes catch curvature and direction-versus-size asymmetry.

Remark 7.1 (Operators sharpen, rather than relax, the multiple-testing problem). Each operator multiplies the column space: one base feature becomes many `op_*` candidates. A `feature_ops` run must therefore log how many columns it minted, so the program-level false-discovery ledger of Section 8 can count them. Minting degrees of freedom and accounting for them are the same obligation.

8 Estimation Hygiene: Keeping the Breadth Estimate Honest

A measured information coefficient or mutual information is an estimate, and estimates of small quantities on short, overlapping, heavy-tailed samples are biased upward by selection. Three processes protect the degrees-of-freedom estimate from being fiction.

Permutation null (reality check). For a candidate set or a prior run, rebuild the null by stationary block bootstrap of returns—which preserves autocorrelation and the overlap structure that naive shuffling destroys (Politis & Romano)—over M resamples, and recompute the headline metric. Report, per feature, $p_{\text{perm}} = M^{-1} \#\{\text{metric}_{\text{null}} \geq \text{metric}_{\text{obs}}\}$, and at the set level the *excess discoveries* $n_{\text{obs}} - \mathbb{E}[n_{\text{null}}]$ (White’s reality check). A set whose excess is ≈ 0 is indistinguishable from luck regardless of any single feature’s p -value.

Stability and decay. Signals are perishable inventory. A non-stationarity audit computes a calendar-windowed (not expanding) IC series, detects breaks by a CUSUM/Page–Hinkley test, and measures sign-consistency and the half-life of $|\text{IC}|$ with window age. A feature is admissible only if no break is detected and its sign is consistent across sub-periods.

Program-level ledger. Per Remark 2.2, the number that decides whether discoveries are real is the *total* count of hypotheses the platform has ever tested. A ledger the runner appends to—(process, target, columns minted, timestamp, git SHA)—is the meta-gate at program scope, the generalization of the reality check from one run to the whole research history. Until it exists, every information number is un-budgeted against the platform’s cumulative search (Bailey, Borwein, López de Prado & Zhu).

9 Verification and Implementation Status

9.1 The planted-signal discipline

No process touches real data before it passes a *planted-signal* conformance test: a synthetic frame with a known signal column and known noise columns, on which the process must flag the signal and must *not* flag the noise. This is not optional hygiene—three estimator bugs

(including the scale artefact of Remark 3.1) were caught only by planted tests, each of which would otherwise have manufactured a discovery. A transform additionally passes a no-lookahead test (a holdout perturbation must not change in-sample outputs). A real-parquet smoke test on the latest day, with dead columns skipped and reasons recorded, is the final pre-commit gate.

9.2 What is built and what is specified

In the interest of honest reporting, Table 4 separates the implemented contract and processes from the specified-but-unbuilt mechanisms. The framework, its result schema, its persistence, and seven processes are implemented; the orthogonalization suite that this paper centres on is specified with its planted contracts and is the immediate build queue.

Implemented	Specified (planted contract written, not yet built)
process contract + result schema + persistence	<code>cmi_select</code> (conditional-MI selection)
<code>ic_horizon</code> (expanding Spearman IC sweep)	<code>interaction_synergy</code> (synergy mining)
<code>mi_ksg</code> (KSG mutual information + CMI)	<code>lead_lag_te</code> (cross-symbol TE)
<code>transfer_entropy</code> (directed information)	<code>residualize</code> (pure innovation)
<code>spectral</code> (PSD / persistence)	<code>permutation_null</code> , <code>stability_decay</code>
<code>ml_importance</code> (walk-forward importance)	<code>regime_conditional</code> , <code>signal_book</code>
<code>triple_barrier</code> , <code>pca_combo</code>	<code>targets</code> registry, <code>feature_ops</code> operators

Table 4: Implementation status. This is a methods/framework paper: the orthogonalization and degrees-of-freedom mathematics are the contribution, the implemented processes demonstrate the contract, and empirical estimation of N_{eff} on a clean multi-day window is specified as future work—meaningful breadth estimates require more independent observations than a thin sample provides.

9.3 The end-to-end pipeline

The processes assemble into a single composition, each arrow a runner chain:

$$\begin{array}{ccccccc}
 \underbrace{\text{feature_ops}}_{\text{mint DOF}} & \rightarrow & \underbrace{\text{cmi_select}}_{\text{pick orthogonal}} & \rightarrow & \underbrace{\text{Target}}_{\text{which price action}} & \rightarrow & \\
 \underbrace{\text{evaluation}}_{\text{score in bits / IC}} & \rightarrow & \underbrace{\text{signal_book}}_{\text{book}+N_{\text{eff}}} & \rightarrow & \underbrace{\text{algorithm}}_{\text{risk-parity combine}} & & .
 \end{array}$$

Two maturity gaps remain explicitly open. The first is execution realism: there is no fill/cost model node, and the fill-conditional target of Table 3 is the hook for the missing fourth citizen, where a mid-price edge becomes a transacted-price edge or does not. The second is the program-level multiple-testing ledger of Section 8. Both are named here precisely so that no estimate downstream is mistaken for more than it is.

10 Conclusion

We have argued that the unit of microstructure research should be the *process*: a typed, registry-bound analytical contract that certifies whether features carry non-redundant information about an explicit target. Around that unit we developed the two pieces of mathematics it exists to serve. Mutual information, justified by Fano’s inequality as a hard floor on achievable error, is the right currency; and the effective degrees of freedom $N_{\text{eff}} = N/[1 + (N - 1)\bar{\rho}]$, derived from the fundamental law of active management, is the right measure of a book’s worth—with the stress-regime correlation, not the calm one, as the binding input. Orthogonal information is then extracted by five composable processes (conditional-MI selection, synergy mining, residualization, denoised principal components, and a capstone signal book), the target is made explicit along an aspect×horizon taxonomy, and new degrees of freedom are minted by a derivative-operator layer under a program-level false-discovery ledger. The result is a single contract under which “does this feature add anything?” and “how many independent bets do we really have?” become routine, testable, reproducible queries rather than one-off scripts. The empirical estimation of N_{eff} on a clean longitudinal window, and the execution-realism node that the fill-conditional target anticipates, are the natural next steps—and, by construction, each will arrive behind its own planted-signal gate.

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